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Lab 5: Indentation

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Overview

Lab 5 focused on measuring how 3D-printed ABS plastic behaves under tiny loads by using micro-indentation. We ran tests on a Nanovea CB500 machine with a Vickers indenter tip, which presses up to 20 N onto the surface without damaging the sample. This gave us detailed measurements of hardness and elastic modulus. Essentially, how the plastic resists and recovers from small dents.

We also tried a hands-on hardness test using a sharpened pencil on soft materials like clay, Jell-O, and Styrofoam. We got a simple, tactile sense of how different materials respond to pressure by feeling how much force it took to leave a mark.

Experimental Setup

We began by selecting a fresh, unmarked region on the 3D-printed ABS specimen and drawing a uniform grid to organize our tests. On the Nanovea CB500 micro-indentation system, we programmed an array of 15 automated indents, each delivered at a constant 4 N load with an unloading rate of 8 N/min. The instrument's integrated microscope was used before every test to zoom in on a pristine spot, ensuring that each indent was made on an untouched surface and that our load-displacement data would be free of artifacts.

For the qualitative portion, we switched to a simple pencil-scale setup. We sharpened a pencil to a fine tip, placed each test material on a zeroed scale, and pressed down slowly until the tip left a visible mark. At that instant, we noted the force reading and later measured the indentation depth so that we could directly compare how easily clay, Jell-O, and Styrofoam gave way under controlled, light pressure.

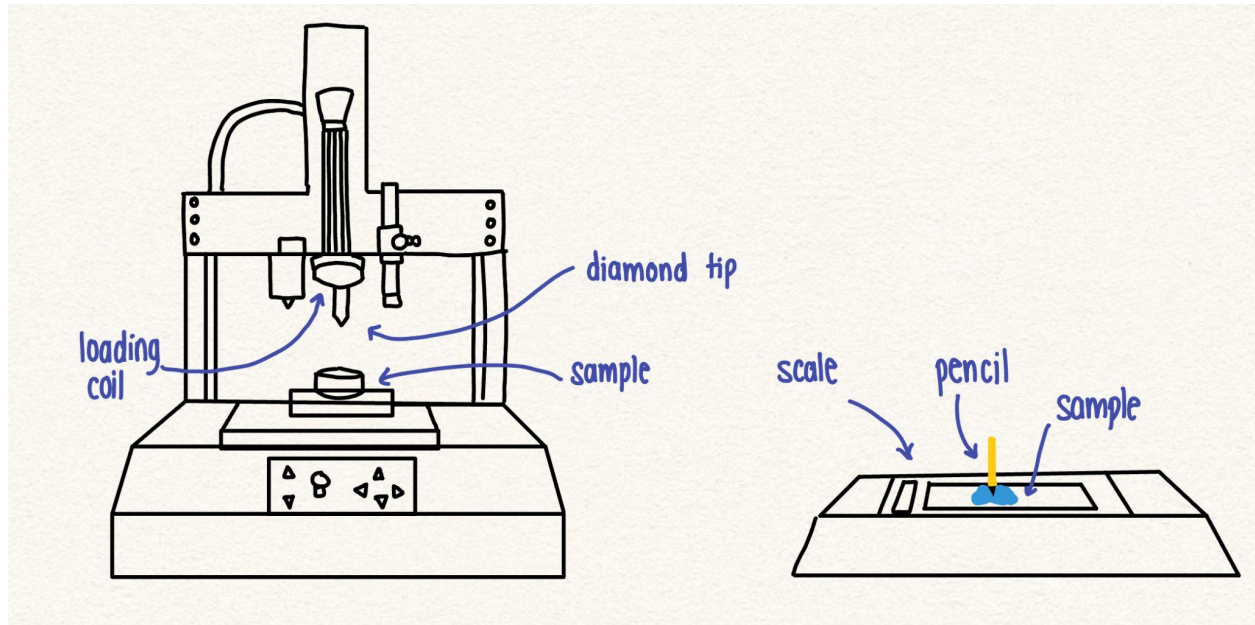


Figure 1 shows two hardness-testing methods: on the left, the Nanovea CB500 applies highly precise, micro-scale indentations to the sample; on the right, a sharpened pencil tip serves as the indenter while a scale records the manual force needed to make each mark.

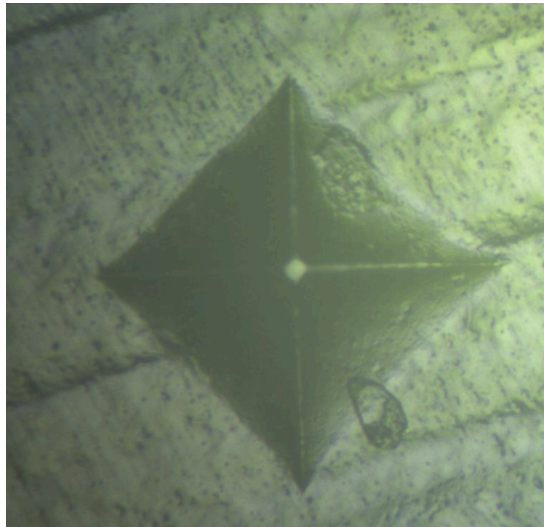


Figure 2 compares two features on the ABS surface: on one side, the CB500's micro-indentation, visible as a smooth, funnel-shaped dent; on the other, a dark, puncture-style mark.

The CB500 indent shows that our test parameters worked as intended, with the diamond tip leaving clear impressions. You'll also see some small porous spots around the indents. These likely come from prior users' tests and reflect normal wear on the machine rather than anything related to our samples.

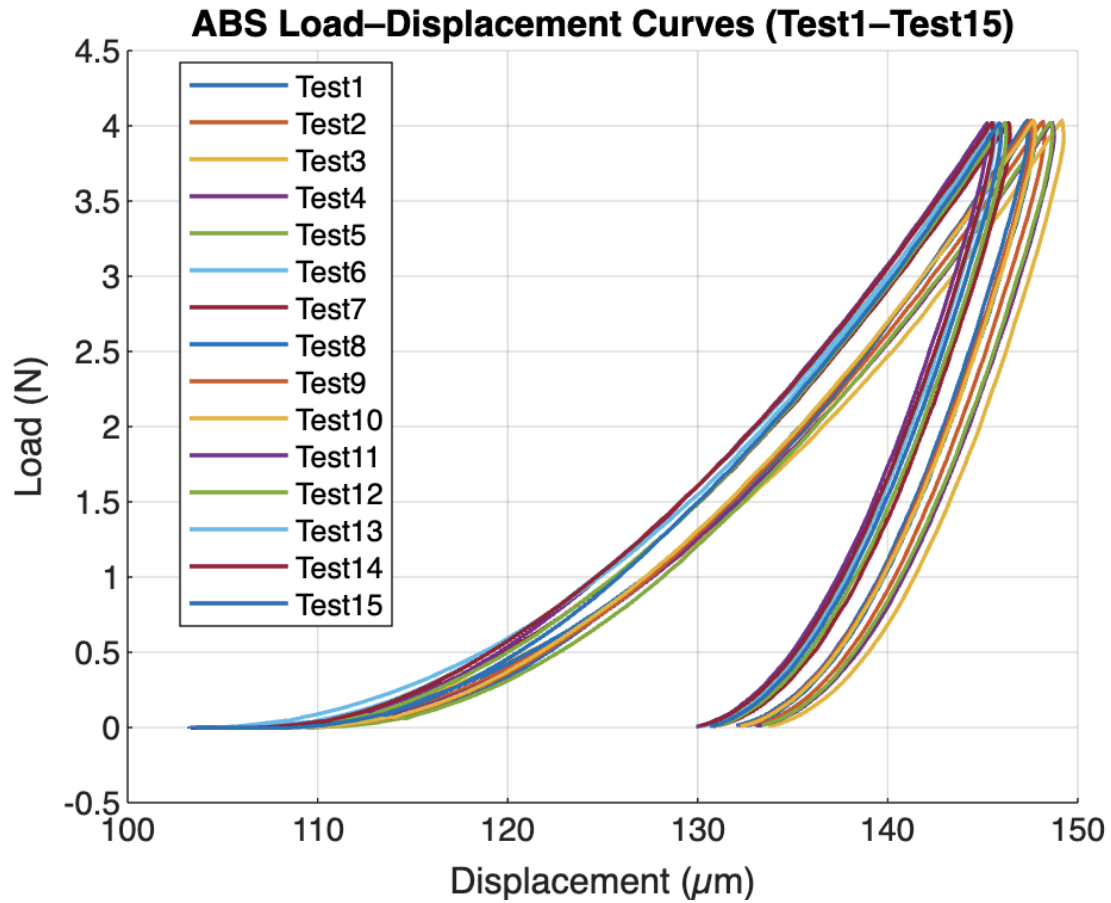


Figure 3 plots indentation depth against applied force for each of the fifteen CB500 trials, with every curve topping out at the 4 N load we set as the maximum

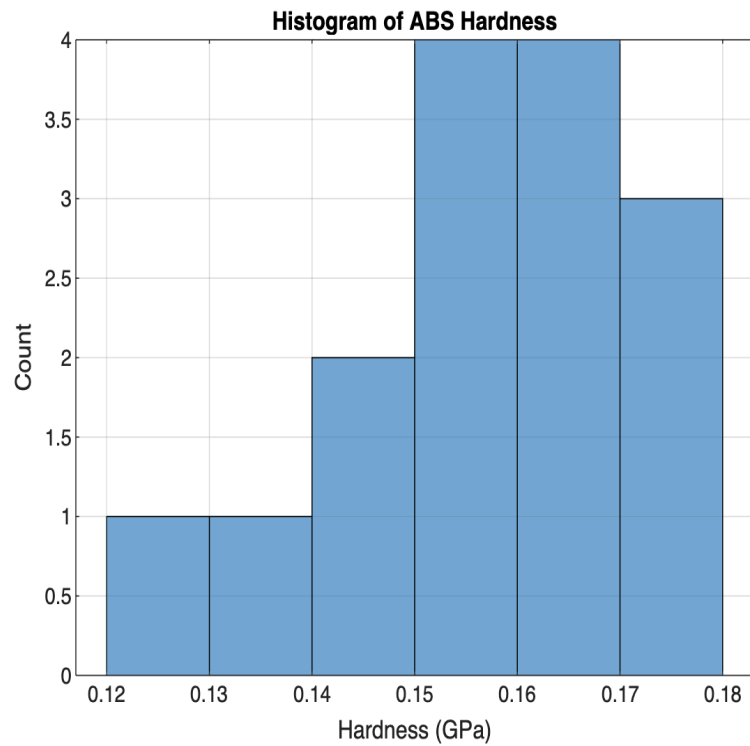


Figure 4: Distribution of hardness values in GPa.

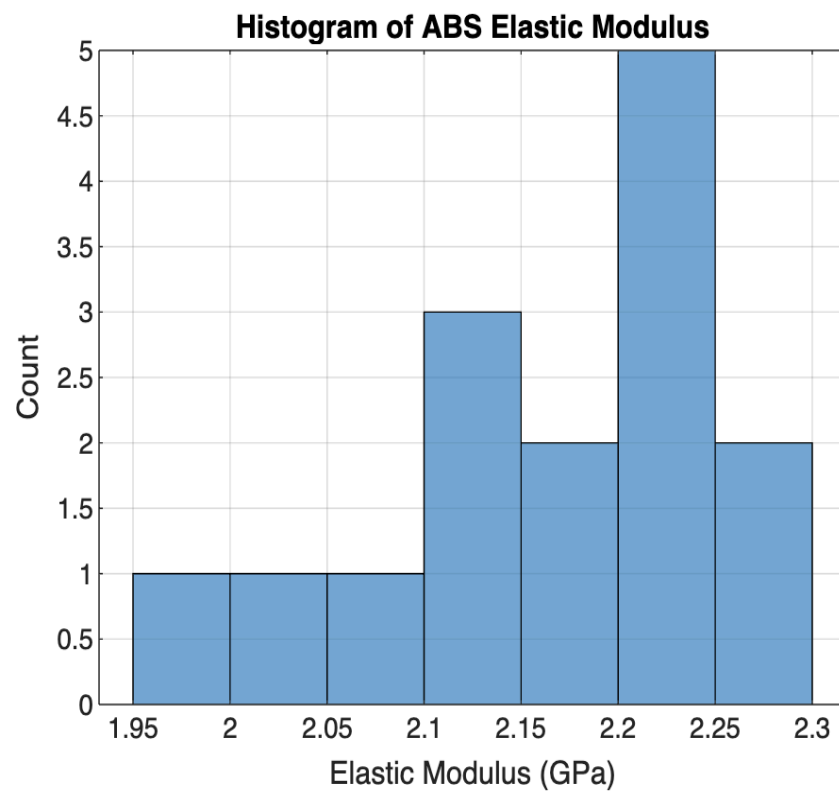


Figure 5: Distribution of elastic modulus values in GPa.

Reported Means and Standard Deviations for the hardness and elastic modulus

By utilizing the standard deviation and mean functions in MATLAB, we found the following:

Hardness:

Mean (GPa): 0.15796

Standard Deviation (GPa): 0.01387

Elastic modulus:

Mean (GPa): 2.16455

Standard Deviation (GPa): 0.0862

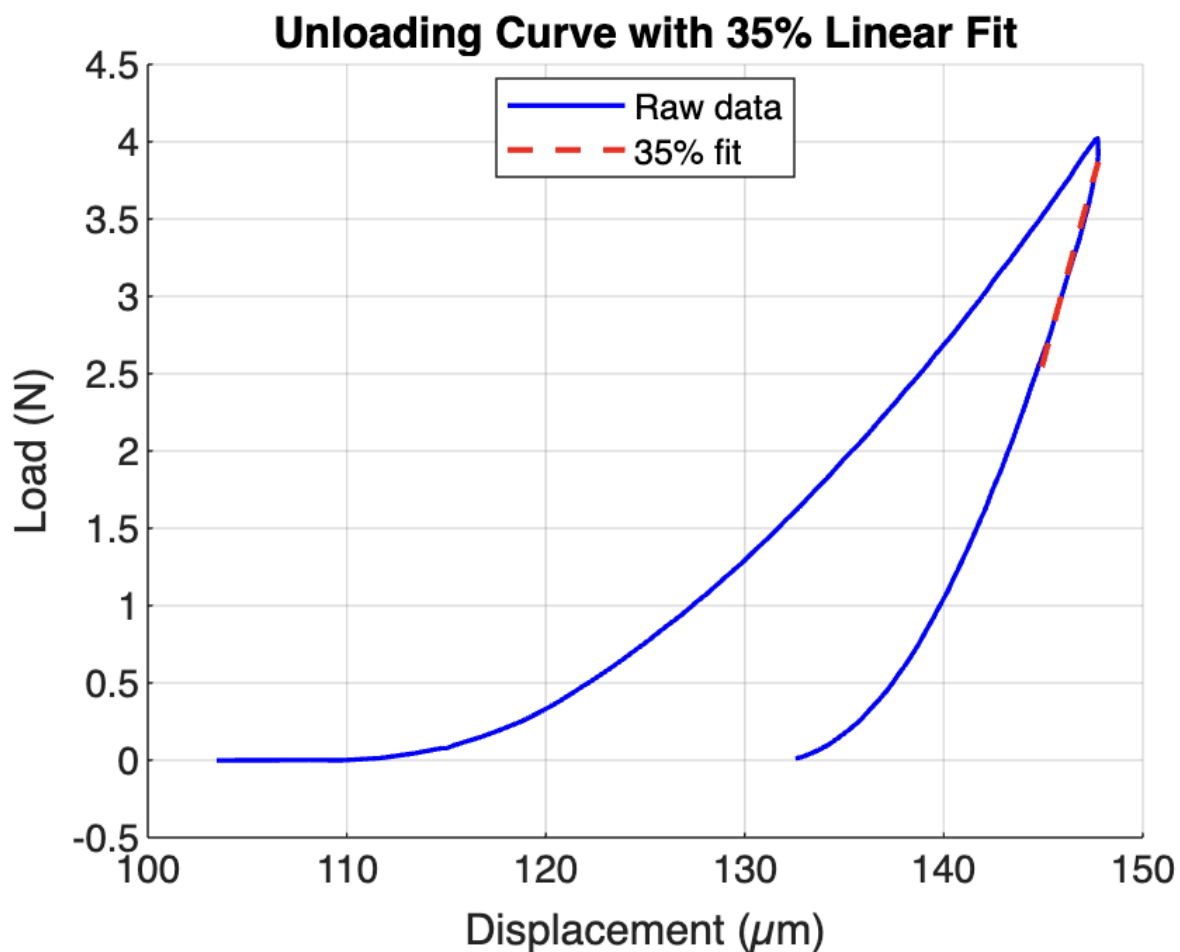


Figure 6 displays the load-displacement curve from the CB500 tests, with a straight-line fit applied to the first 35 % of the unloading segment to determine stiffness. From that fit, MATLAB calculated a reduced modulus (E_r) of 2.47 GPa and derived the ABS elastic modulus as 2.82 GPa and found a Vickers hardness of 14.65 kgf/mm².

Hardness and Modulus calculations

Hardness : $H = F / A$ $F = 4.022 \text{ N}$

$A = d^2 / 2 \sin(136^\circ / 2)$ $h = 46.24 \text{ } \mu\text{m}$

$d = 2 h \tan(136^\circ / 2) = 2 (46.24) \tan(68^\circ) = 0.228 \text{ mm}$

$A = (0.228)^2 / 2 \sin(68^\circ) = 0.028 \text{ mm}^2$

$H = 4.022 \text{ N} / 0.028 = 0.41045 \text{ kgf} / 0.028 \text{ mm}^2 =$

$= 14.65 \text{ kgf} / \text{mm}^2$

Modulus

$S = \frac{2}{\sqrt{\pi}} E_r \sqrt{A}$ $E_r = \frac{S \sqrt{\pi}}{2 \sqrt{A}} = \frac{(4.67 \times 10^5) \sqrt{\pi}}{2 (2.8 \times 10^{-8})} = 2.47 \text{ GPa}$

$S = 4.67 \times 10^5 \text{ N/m}$ $A = 2.8 \times 10^{-8} \text{ m}^2$

$E = E_r / (1 - \nu^2) = (2.47 \times 10^9) / (1 - 0.35^2) = 2.82 \text{ GPa}$

Figure 7 shows calculations for Hardness and Modulus as well as calculations for the area of the indent, which results in being 0.028mm².

Possible Causes of Variation in Hardness and Modulus Data

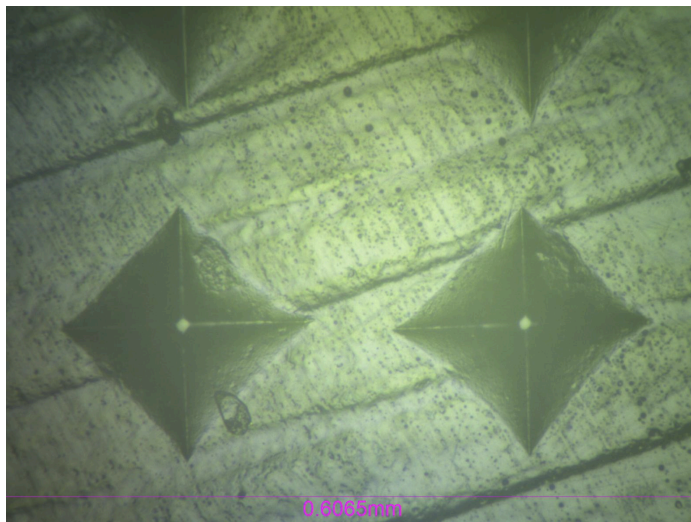


Figure 8: This image highlights two sample indents. Examining the load-displacement traces from all fifteen trials reveals some variability—Test 1’s hardness is 0.01 GPa below the mean, and its elastic modulus is 0.31 GPa lower than the average. These offsets likely stem from slight inconsistencies in the ABS—visible streaks and texture differences in the photo—and the minor differences in indent geometry. Despite these small shifts, every curve still follows the ideal profile, showing that the variations are inconsequential overall.

Determining if Measured Values are Significantly Different from Expected Values

Hardness: $t(14)=2.222$, $p=0.0432 \rightarrow \text{reject } H_0$

Modulus: $t(14)=7.391$, $p=3.3966e-06 \rightarrow \text{reject } H_0$

Figure 9: Results from Student’s t-tests comparing our measured ABS properties to their expected literature values. The Vickers hardness (mean 0.1579 GPa) falls within a 95 % confidence interval of 0.1503–0.1656 GPa, while the elastic modulus (mean 2.1645 GPa) lies between 2.1168–2.2123 GPa. Testing against theoretical benchmarks (0.15 GPa for hardness and 2.0 GPa for modulus) gives $t = 2.222$ ($p = 0.0423 < 0.05$) and $t = 7.391$ ($p < 0.00001$), respectively, indicating statistically significant differences likely caused by slight sample inconsistencies.

Calculating The Area vs Depth of the Pencil Indent

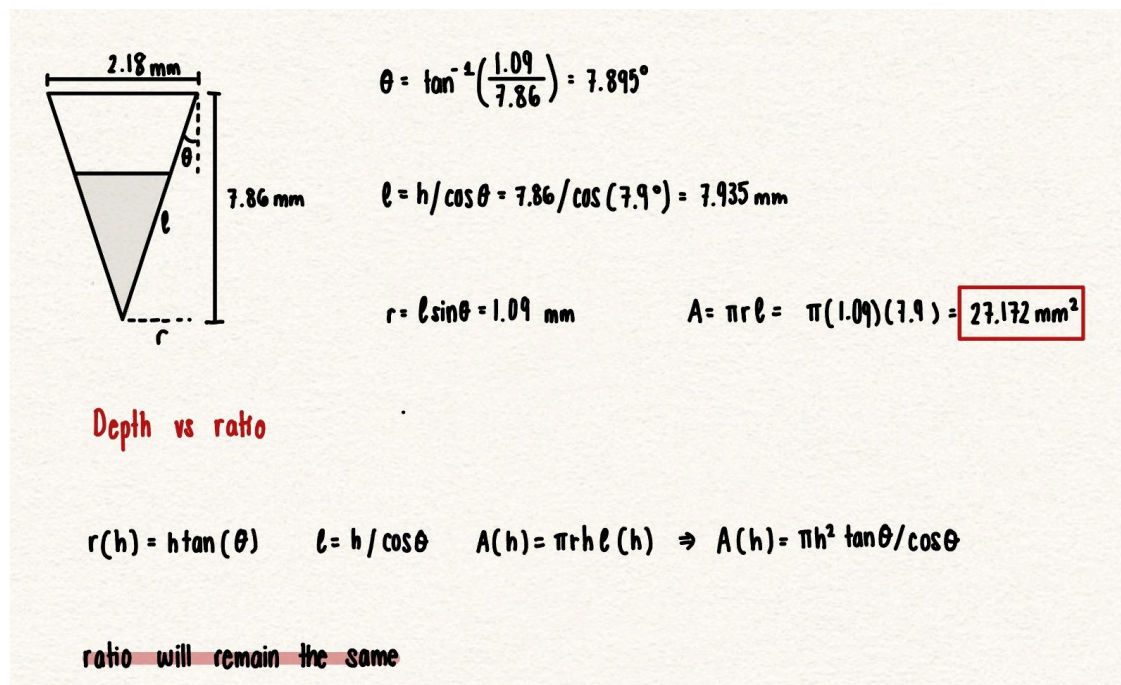


Figure 10: Depicts the indenter's full engagement with the sample, showing the actual contact footprint when the tip is fully embedded. It also shows how that contact area grows as penetration depth increases, clarifying the direct relationship between depth and contact size. The are was found to be 27.19 mm².

Reporting Measured Load, Depth, and Diameter of Indent from Pencil on Three Different Materials.

Table 1: Qualitative Hardness Measurements

Material	Load (g)	Force (N)	Depth of Indent (mm)	Width of Indent (mm)	Hardness from Indent Depth (GPa)	Hardness from Indent Width (GPa)
Styrofoam	550	5.39	7.86	2.18	0.000198	0.000198
	600	5.88	7.86	2.18	0.000216	0.000216
	575	5.64	7.86	2.18	0.000207	0.000207
Clay	500	4.90	7.86	2.18	0.000180	0.000180
	540	5.30	7.86	2.18	0.000194	0.000194
	510	5.00	7.86	2.18	0.000184	0.000184
Jell-O	2	0.0196	7.86	2.18	7.20e-7	7.20e-7
	3	0.0294	7.86	2.18	1.08e-7	1.08e-7
	4	0.0392	7.86	2.18	1.44e-6	1.44e-6

Table 1: Manual hardness tests on Styrofoam, clay, and Jell-O using a sharpened pencil. For each material, we recorded the weight on a scale (converted to force), measured the indentation depth and width, and calculated hardness from both dimensions. Because we (incorrectly) held the indent diameter constant with a fixed depth, both depth-based and width-based hardness values came out the same. In future trials, factors like material pile-up around the indent, rather than brittle fracture, should be considered as potential sources of variation.

Photos of Indents Made by Pencil on all Three Different Materials



Figure 11: Styrofoam indentations. The styrofoam sample after testing, showing three distinct impression marks made during the hardness measurements.

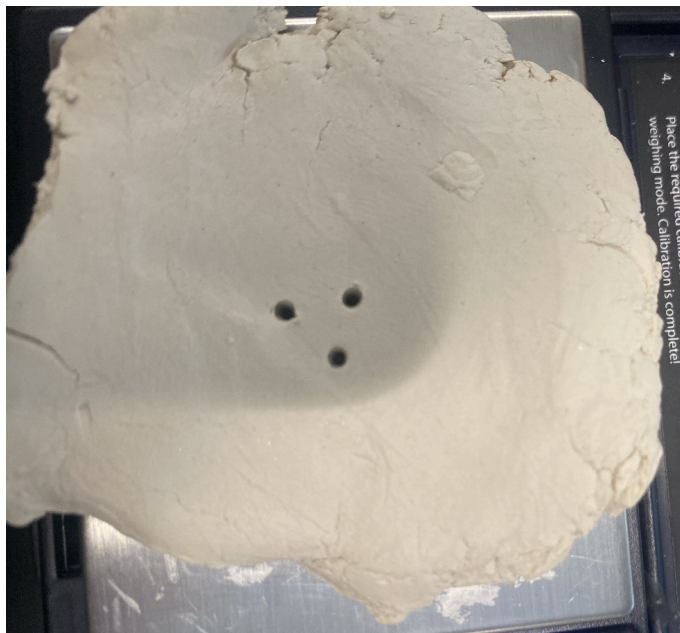


Figure 12: Clay Indentations, showing three clearly defined indent marks from the hardness measurements.



Figure 13: Jello Indentations, showing three clearly defined indent marks from the hardness measurements.. Note that these indentations are subtle and may be difficult to see.

Comparison of Material Hardness: Sources vs. Our Tests

We looked up published hardness values to compare with our tests. According to MatWeb, Styrofoam has a Vickers hardness between 0.076 and 0.15 GPa. Clay hardness can vary widely. Mudstone clay averages about 0.5 ± 0.5 GPa, but because we used soft, oil-based modeling clay rather than fired clay, its real hardness is expected to be much lower. Gelatin gels like Jell-O are extremely soft, with reported hardness on the order of 10^{-5} GPa (roughly $9\text{--}13 \times 10^{-6}$ GPa) in the biomedical literature

Discussion:

Our pencil-scale tests for Styrofoam and clay came in much softer than the published values in Table 1. There are a few likely reasons for this gap. The indenter setup itself may have calibration issues that caused us to under-report the actual force applied. Because these materials barely resist indentation, it was also easy to lift the pencil too soon and record the scale reading before reaching full load. And since our Styrofoam came from a thin, previously used sheet (Figure 10), it flexed and yielded far more under pressure than a fresh, thicker sample would.